SecureWave

User Manual



Secure Text | Secure Files | Secure Radio | Secure Pairing

About this manual: Screenshots use test data. Protected screens that cannot be captured are shown as clearly marked illustrative reconstructions. SecureWave is a PUBLIC BETA and has not been independently security-audited or certified. Screenshots may show an earlier interface; follow the labels on your installed version.

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1. Quick Start

For a new installation, the recommended sequence is:

1. Open Contacts and create or securely pair a contact.

2. Prefer Hybrid Secure Pairing when practical. Compare every group of the verification code before saving trust.

3. Open Secure Text, select the contact, type a short test message and send it.

4. If using radio, verify RX, audio routing and PTT before sending an important message.

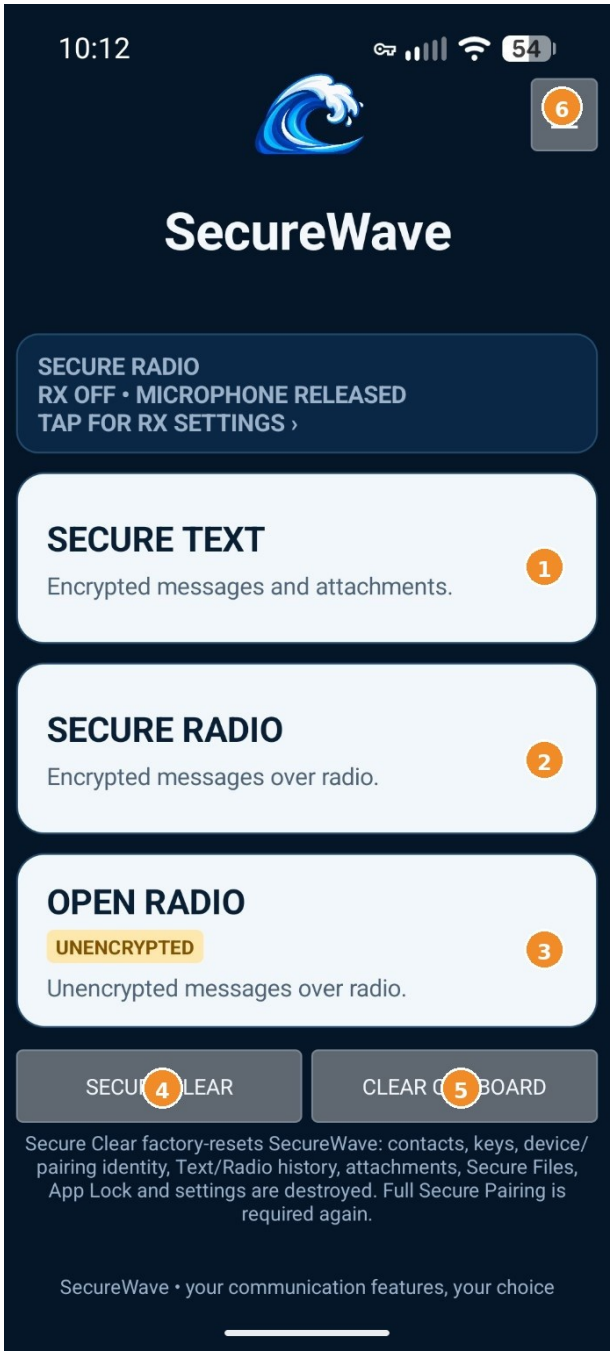
5. Use Secure Files for encrypted local storage and for importing encrypted file transports.

Important: Open Radio is explicitly unencrypted. Use Secure Radio for encrypted radio messages to a secure contact.

Verification: A pairing exchange is not trusted merely because it completed. Trust is committed only after both users confirm that the complete verification code matches.

2. Home

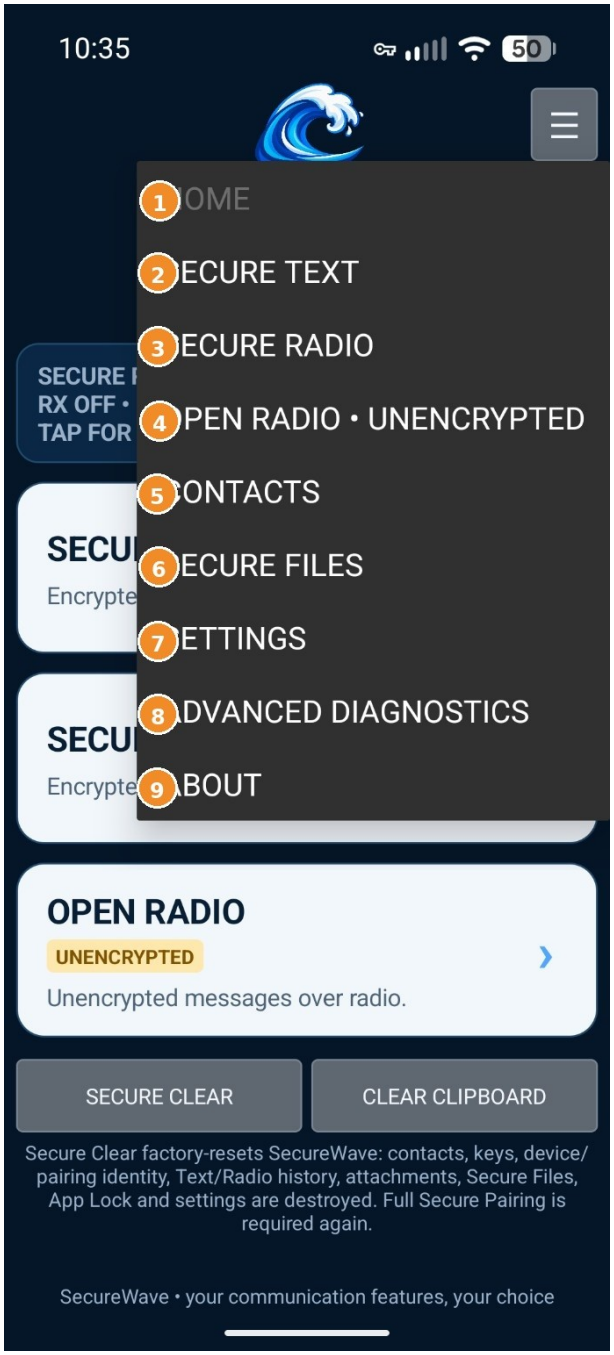
The Home page provides the primary communication modes and quick security controls.



1. SECURE TEXT opens encrypted conversations and attachments.
2. SECURE RADIO opens encrypted radio messaging for a selected secure contact.
3. OPEN RADIO is unencrypted and is marked accordingly.
4. SECURE CLEAR destroys protected SecureWave state; use only intentionally.
5. CLEAR CLIPBOARD clears SecureWave-copied plaintext when applicable.
6. The three-bar menu opens global navigation.

2.1 Global Navigation

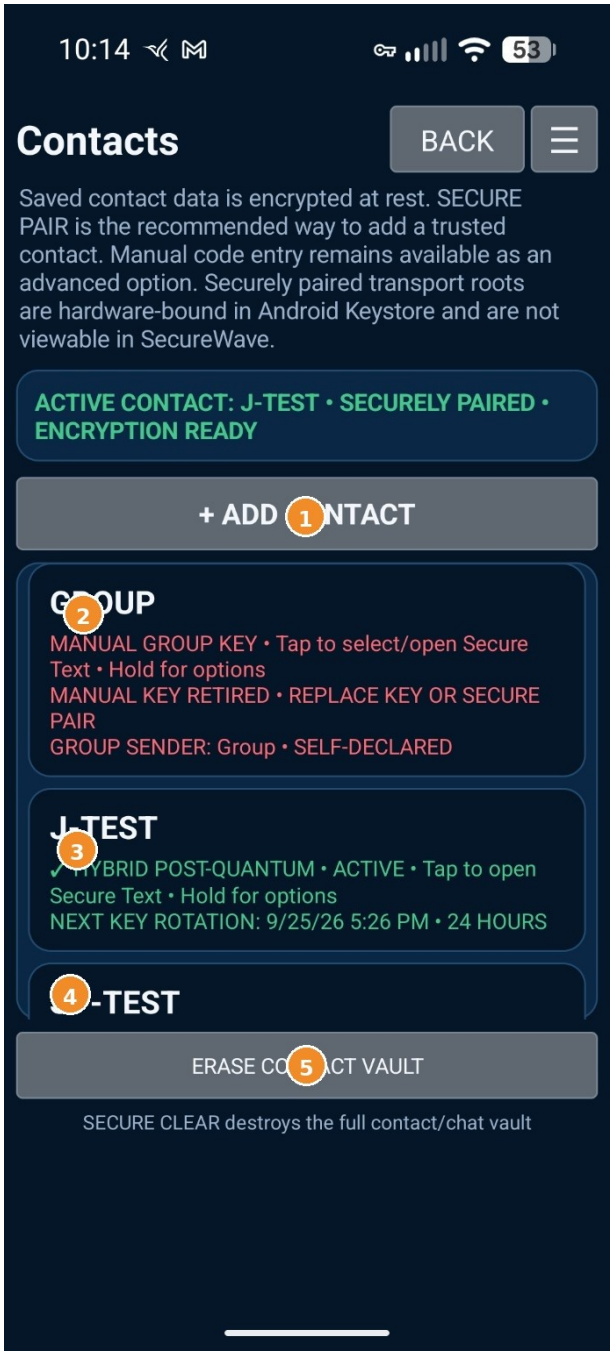
Use the three-bar menu to move between major SecureWave areas.



1. Home.
2. Secure Text.
3. Secure Radio.
4. Open Radio - explicitly unencrypted.
5. Contacts.
6. Secure Files.
7. Settings.
8. Advanced Diagnostics.
9. About / build information.

3. Contacts

Contacts stores manual and securely paired relationships, encrypted at rest. Current public beta builds include live contact search and Favorites. The screenshot below illustrates contact security states; its layout may differ from the installed version.



1. ADD CONTACT creates a manual contact or starts a secure pairing flow.

2. A manual shared-key group is visibly distinguished and sender identity is self-declared.

3. A Hybrid post-quantum paired contact shows its active secure status and rotation information.

4. Use Search contacts and Favorites to find a recipient; use the visible contact options control for editing and other actions.

5. ERASE CONTACT VAULT removes the contact/chat vault.
Secure Clear is broader and destroys additional state.

3.1 Current Contact Navigation and Manual Keys

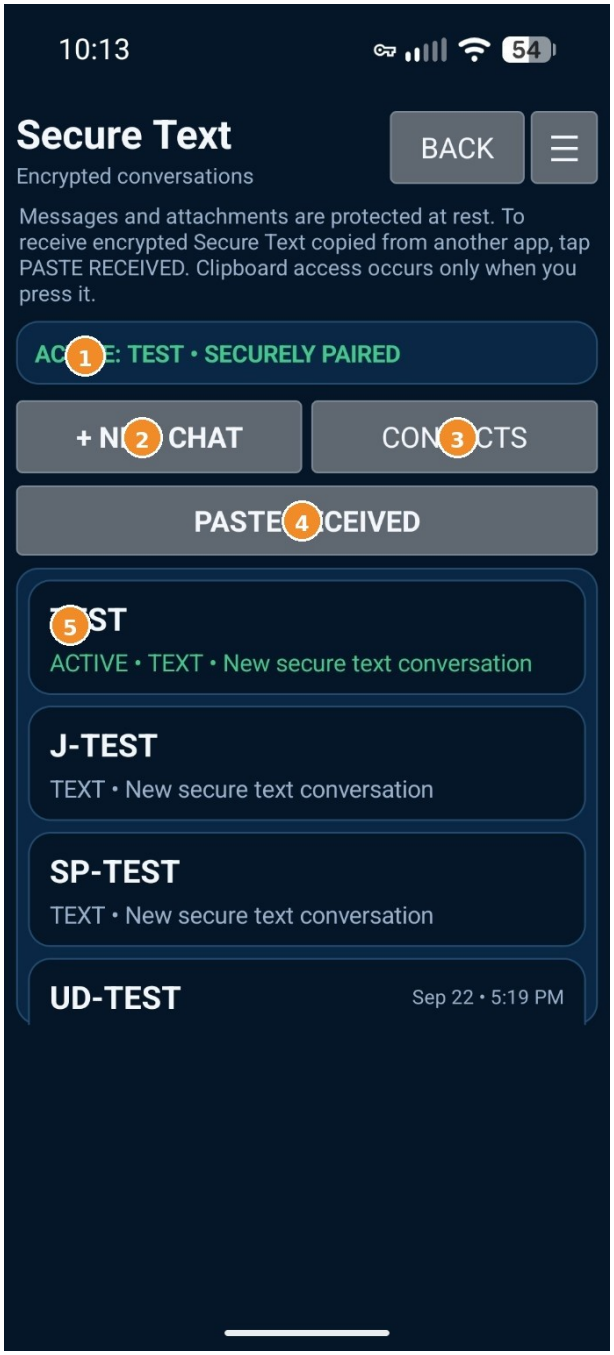
Use Search contacts to filter by any part of a name (case/accent insensitive). Star commonly used contacts to display Favorites first. Search results and Favorites do not change the active recipient until you deliberately select a contact.

For a manual contact, generate a secure 32-character code or enter the shared code. COPY CODE becomes available only when the code is valid; copying is deliberate, and the SecureWave clipboard-clear target is no more than 30 seconds. Android background restrictions mean clearing is best-effort. Share this secret only through a trusted channel.

Manual shared-key groups do not authenticate individual sender names: every participant holding the group key can claim a name. Prefer verified Hybrid Secure Pairing when practical.

4. Secure Text - Conversations

The Secure Text list shows conversations. The recipient explicitly imports encrypted payloads using PASTE & DECRYPT; SecureWave does not send messages directly over the Internet.



1. ACTIVE identifies the currently selected contact; always verify the recipient before sending.

2. CHATS / NEW CHAT navigates to conversations or begins a new one (labels vary by screen).

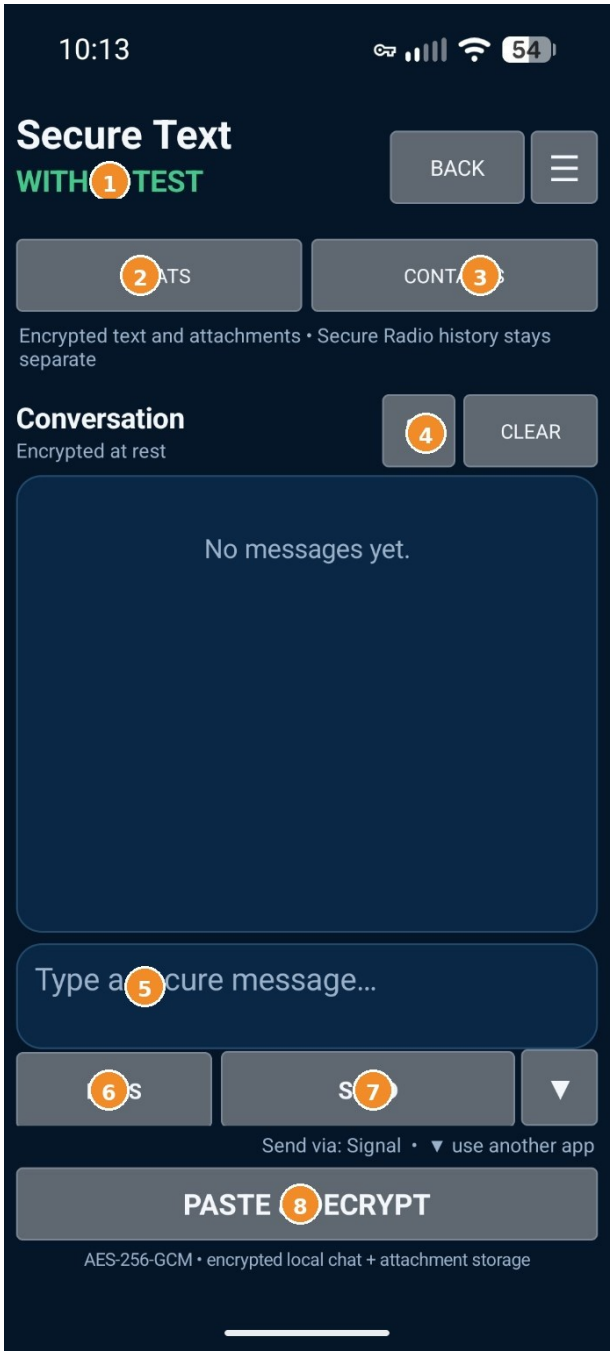
3. CONTACTS opens contact management.

4. PASTE & DECRYPT imports an encrypted received payload only when you request it.

5. Conversation cards show contact and conversation state.

4.1 Secure Text - Conversation

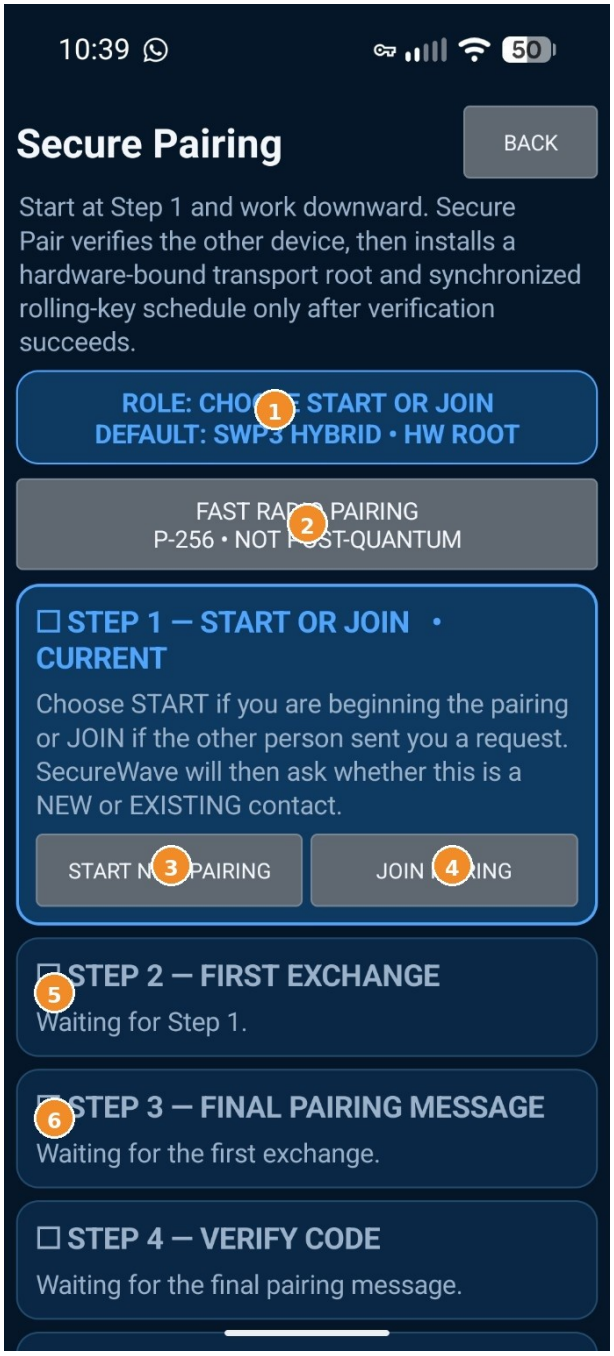
Secure Text protects message content and attachments for the selected contact. Confirm the displayed recipient before sending; the external sharing app handles transport.



1. WITH identifies the selected contact.
2. CHATS returns to the conversation list.
3. CONTACTS opens contact selection/management.
4. The Conversation pane contains encrypted-at-rest history.
5. The message composer grows and then scrolls for longer drafts.
6. FILES attaches encrypted files.
7. SEND encrypts and hands off the message using the selected transport.
8. PASTE & DECRYPT imports encrypted text received through another app.

5. Hybrid Secure Pairing - Start

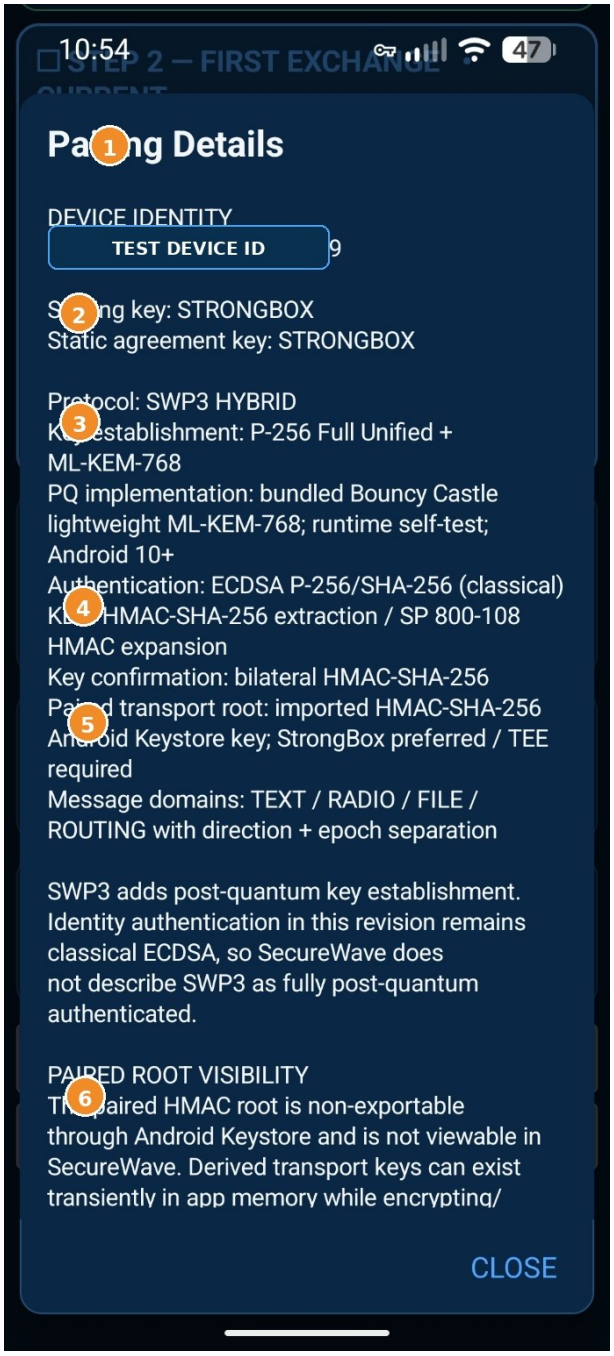
Hybrid Secure Pairing is the recommended pairing method. It combines P-256 and ML-KEM key establishment with a mandatory human verification step.



1. The role banner shows whether you are starting or joining and identifies the default Hybrid pairing path.
2. FAST RADIO PAIRING is a separate P-256-only option and is not post-quantum.
3. START NEW PAIRING begins a new Hybrid pairing session.
4. JOIN PAIRING imports the other user's request.
5. Step 2 becomes available after Step 1 completes.
6. Step 3 carries the final pairing message before verification.

5.1 Pairing Details

Pairing Details explains the cryptographic and hardware backing of the current Hybrid pairing implementation.



1. Device identity fingerprint (shown here as test data).

2. Persistent signing and static agreement keys are hardware-backed when supported.

3. The protocol combines P-256 and ML-KEM-768 for key establishment.

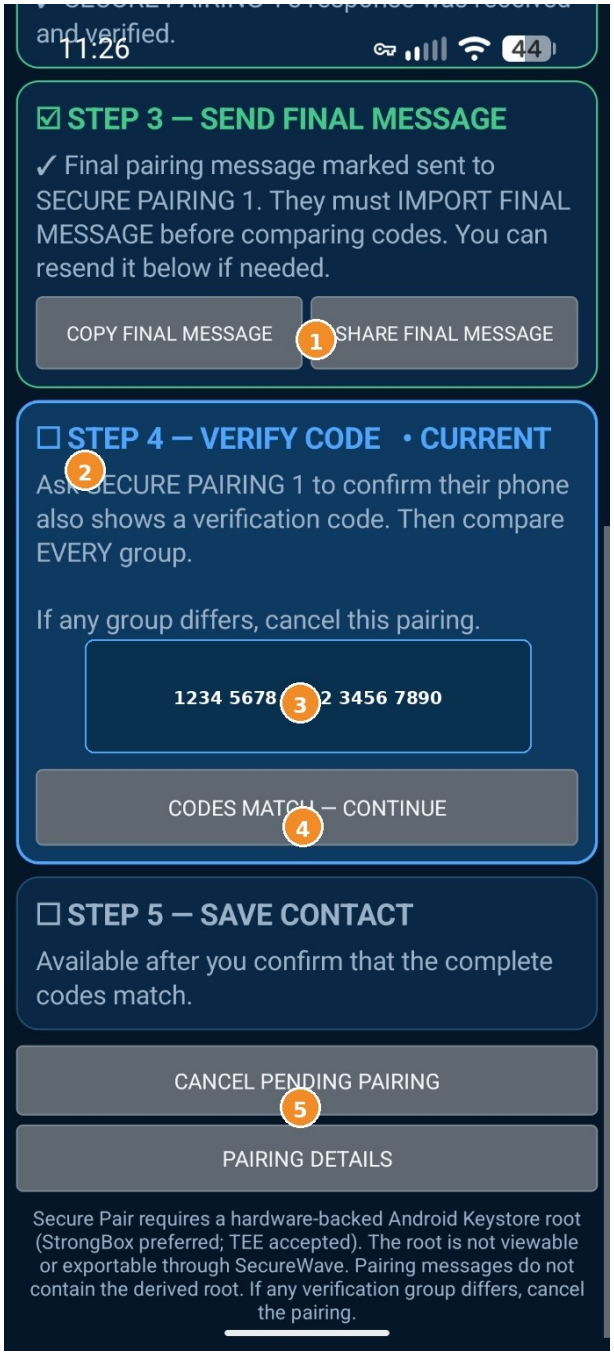
4. Authentication is ECDSA P-256/SHA-256 in this revision.

5. Key confirmation and KDF details describe how both sides verify the established secret.

6. The hardware-backed paired root is non-exportable; TEXT / RADIO / FILE / ROUTING are separated.

5.2 Verify the Hybrid Pairing Code

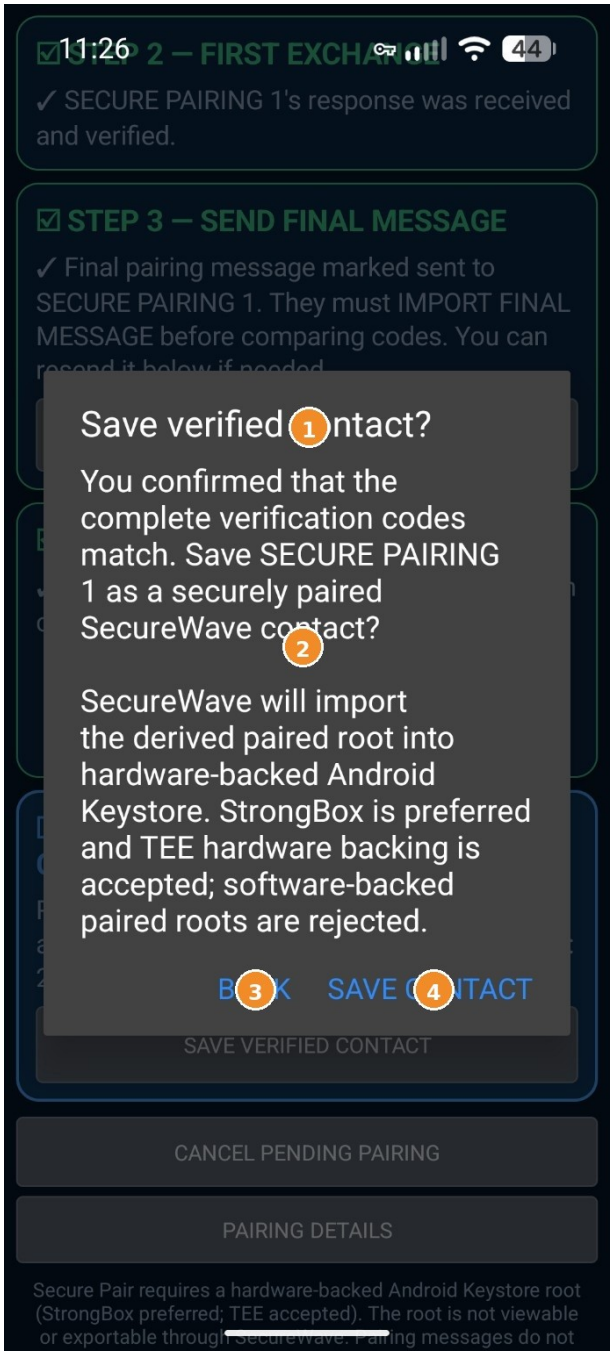
After the exchange completes, both users compare the entire 20-digit code before saving the contact.



1. Step 3 must be complete and the final pairing message must reach the other device.
2. Step 4 is the active verification step.
3. Compare every group of the code. The manual uses a fictional example code.
4. CODES MATCH - CONTINUE advances only after you personally confirm the match.
5. CANCEL PENDING PAIRING is the correct action if any group differs.

5.3 Save the Verified Contact

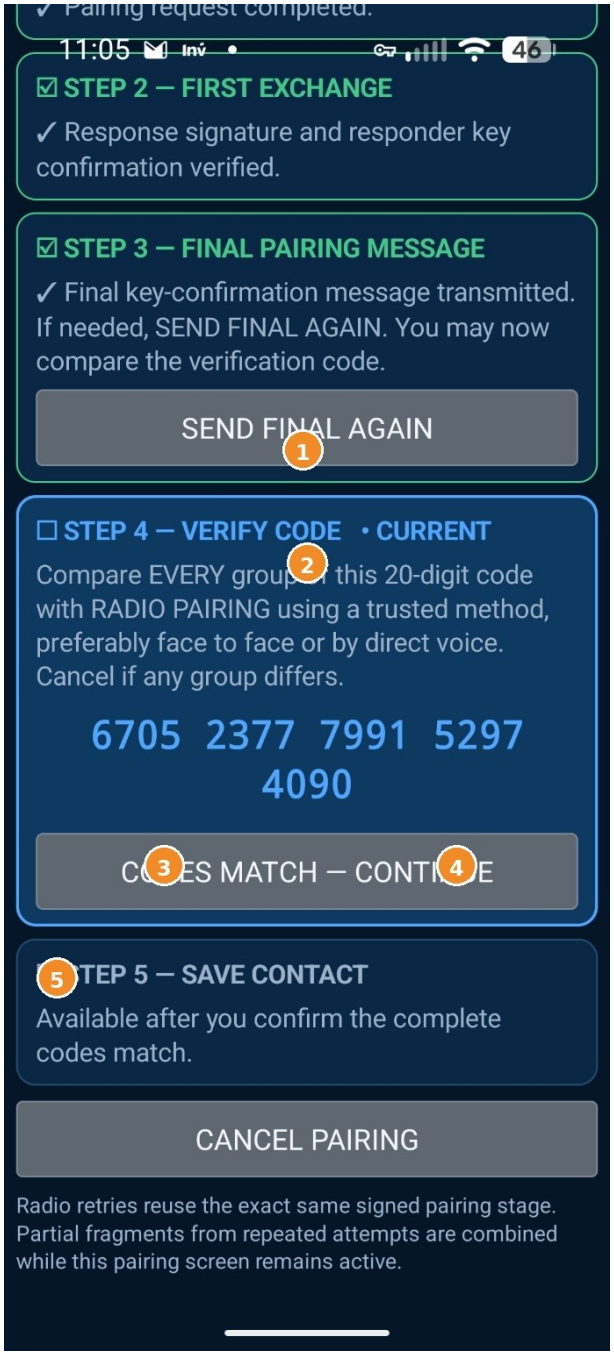
SecureWave asks for one final confirmation before the verified relationship is committed.



1. The confirmation reminds you that the complete verification codes must already have been compared.
2. The derived paired root is imported into hardware-backed Android Keystore protection; StrongBox is preferred and TEE is accepted.
3. BACK leaves without saving the relationship.
4. SAVE CONTACT commits the verified contact.

6. Fast Radio Pairing - Start

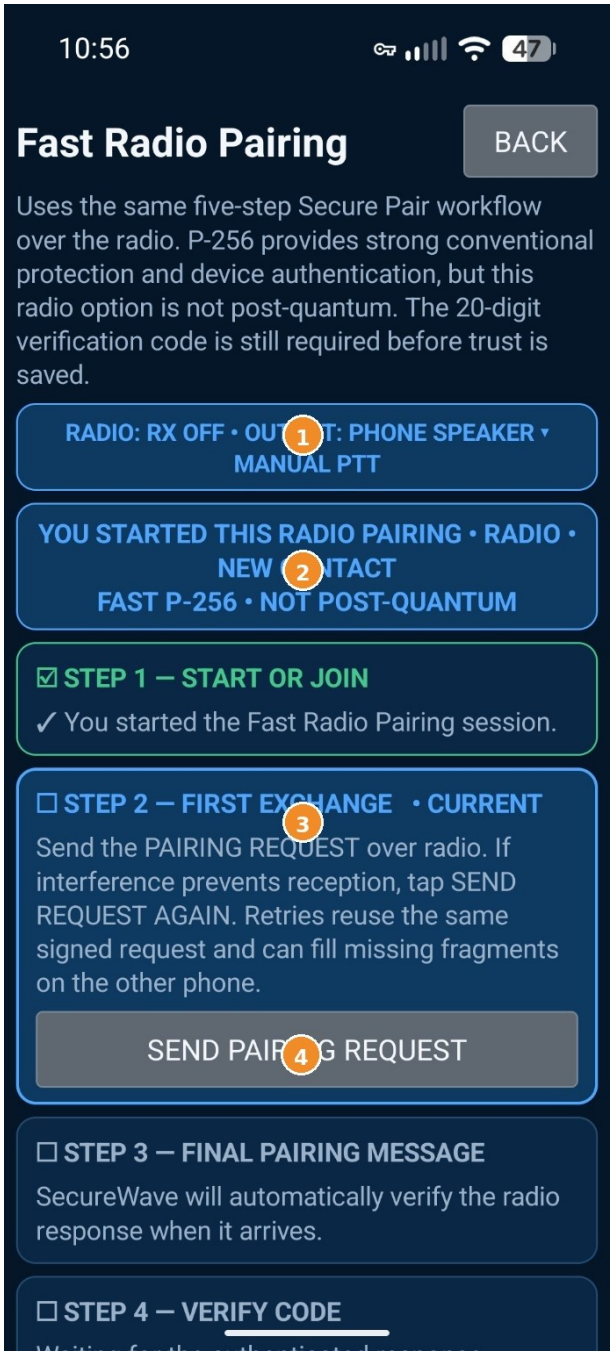
Fast Radio Pairing uses the same five-step mental model over radio, but uses P-256 and is explicitly not post-quantum.



1. Radio readiness shows RX state, output route and PTT mode.
2. The role banner identifies Fast P-256 Radio and its non-PQ status.
3. START RADIO PAIRING begins the request.
4. JOIN RADIO PAIRING waits for the other user's request.
5. Later steps remain locked until the earlier exchange completes.

6.1 Fast Radio Pairing - Request and Retry

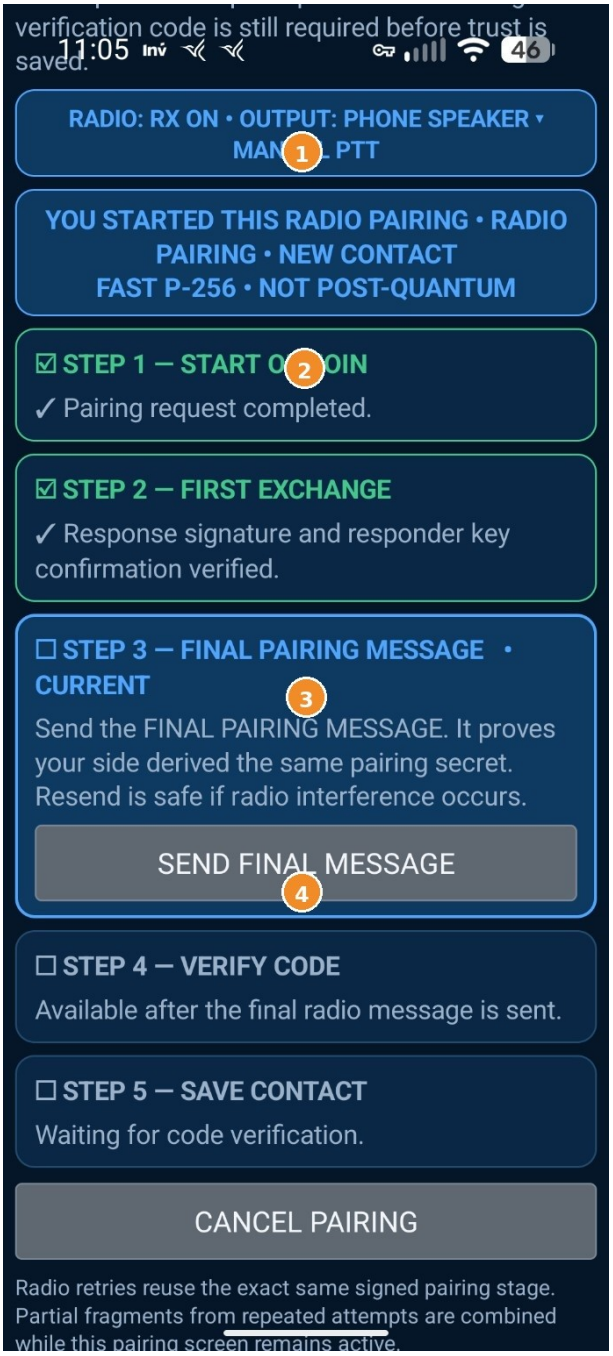
Radio interference is expected to be recoverable without restarting the pairing session.



1. The session banner confirms who started the pairing and the security mode.
2. Step 1 is complete.
3. Step 2 sends the signed pairing request over radio.
4. If reception is incomplete, SEND REQUEST AGAIN retransmits the same signed request so missing fragments can be recovered.

6.2 Fast Radio Pairing - Final Message

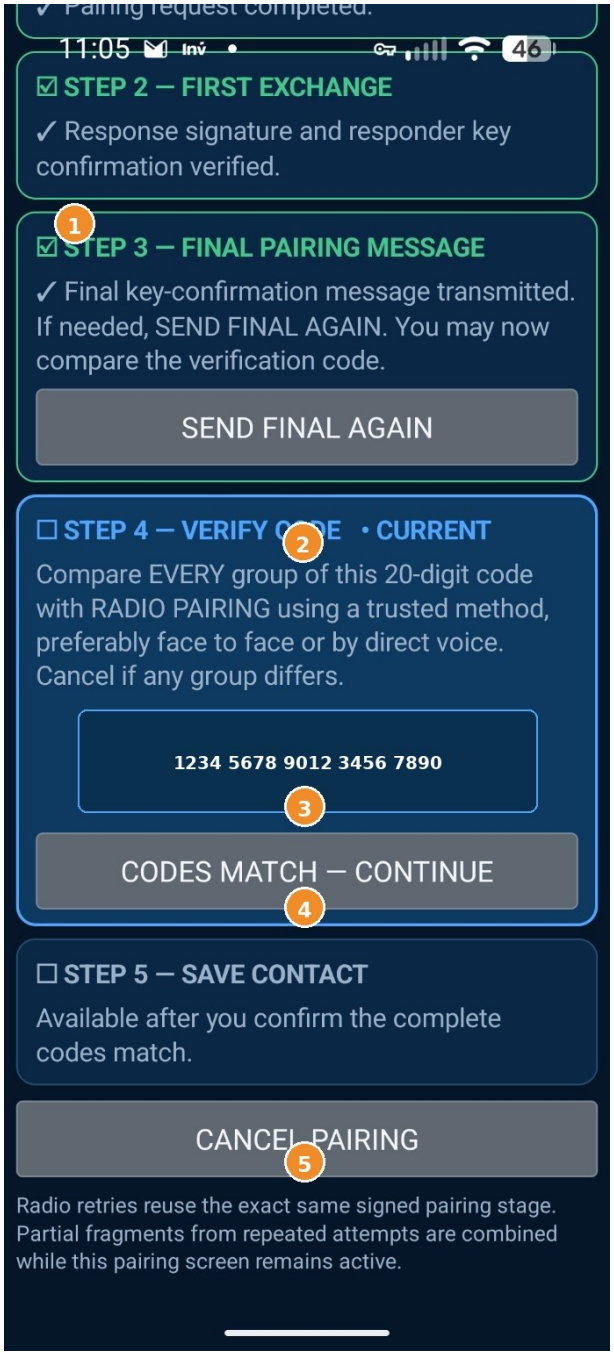
After the signed response and responder key confirmation verify, the initiator sends the final key-confirmation message.



1. Completed steps remain visibly marked.
2. The response signature/key confirmation has verified.
3. SEND FINAL MESSAGE transmits the final confirmation.
4. Verification becomes available only after the final radio message is sent.

6.3 Fast Radio Pairing - Verify and Resend

The final stage can be resent, and the same mandatory verification-code rule applies.



1. SEND FINAL AGAIN repeats the same authenticated final stage if needed.
2. Step 4 is the current verification stage.
3. Compare every group of the displayed 20-digit code; the manual shows fictional example digits.
4. CODES MATCH - CONTINUE proceeds only after a complete match.
5. CANCEL PAIRING abandons the pending relationship.

7. Secure Files - Empty State

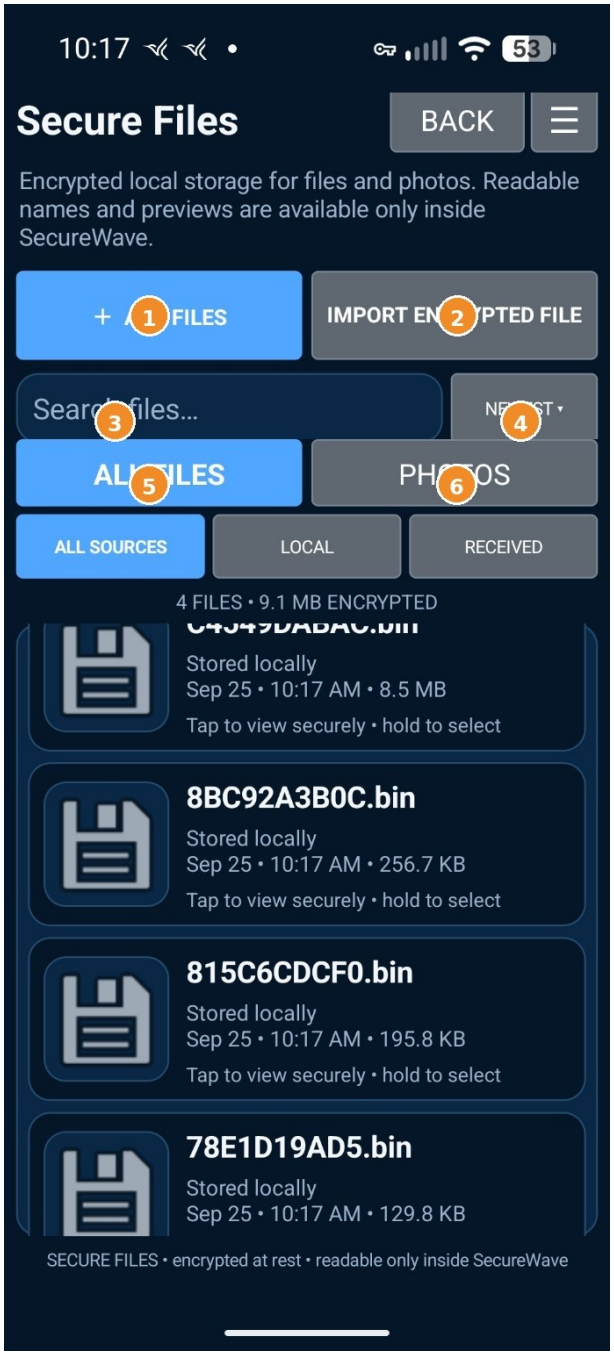
Secure Files is encrypted local storage for files and photos, with explicit import of received encrypted file transports.



1. ADD FILES imports local files into the encrypted vault.
2. IMPORT ENCRYPTED FILE imports a received opaque encrypted transport.
3. The same Add action is repeated in the empty-state card.
4. The same Import action is repeated for convenience.
5. SECURITY DETAILS explains vault and screenshot behavior.

7.1 Secure Files - Search and Filters

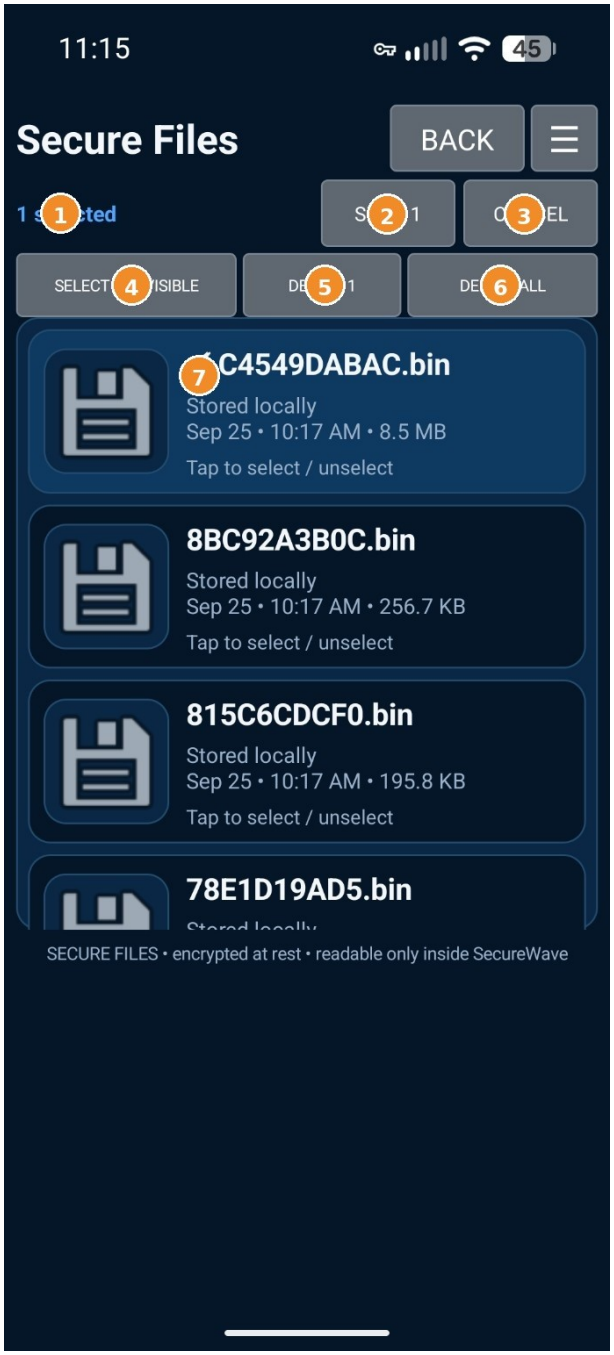
When files are present, Secure Files adds search, sort and source/type filters.



1. ADD FILES.
2. IMPORT ENCRYPTED FILE.
3. Search by visible SecureWave file name.
4. Sort order.
5. ALL FILES / PHOTOS type filter.
6. ALL SOURCES / LOCAL / RECEIVED source filter.

7.2 Secure Files - Select, Send and Delete

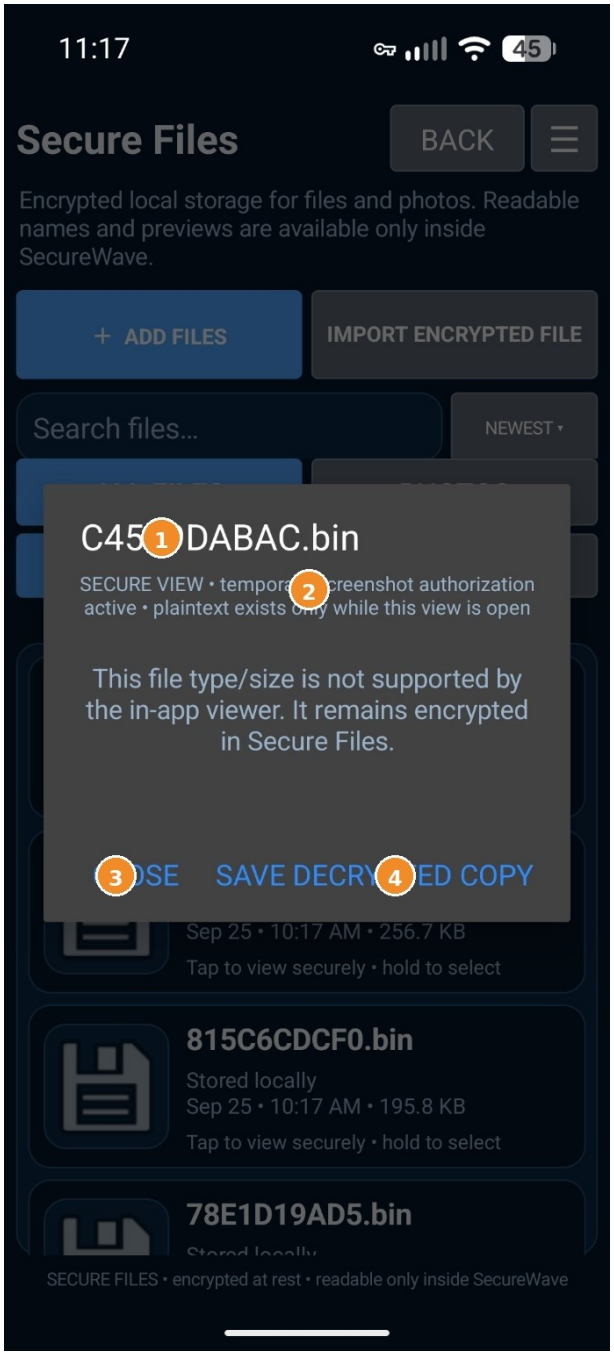
Long-press/select mode supports sending and deleting one or more stored items.



1. Selected item count.
2. SEND 1 sends the selected encrypted item.
3. CANCEL exits selection mode.
4. SELECT ALL VISIBLE selects the current filtered view.
5. DELETE 1 removes the selected item.
6. DELETE ALL removes all visible matching items.
7. A checkmark identifies the selected card.

7.3 Secure Files - View / Export

Opening a stored file keeps it inside a protected Secure View when supported.



1. The file name is shown inside SecureWave.
2. SECURE VIEW notes that plaintext exists only while the view is open.
3. CLOSE returns without exporting plaintext.
4. SAVE DECRYPTED COPY intentionally writes a plaintext copy outside the SecureWave vault.

7.4 Large Encrypted Attachments

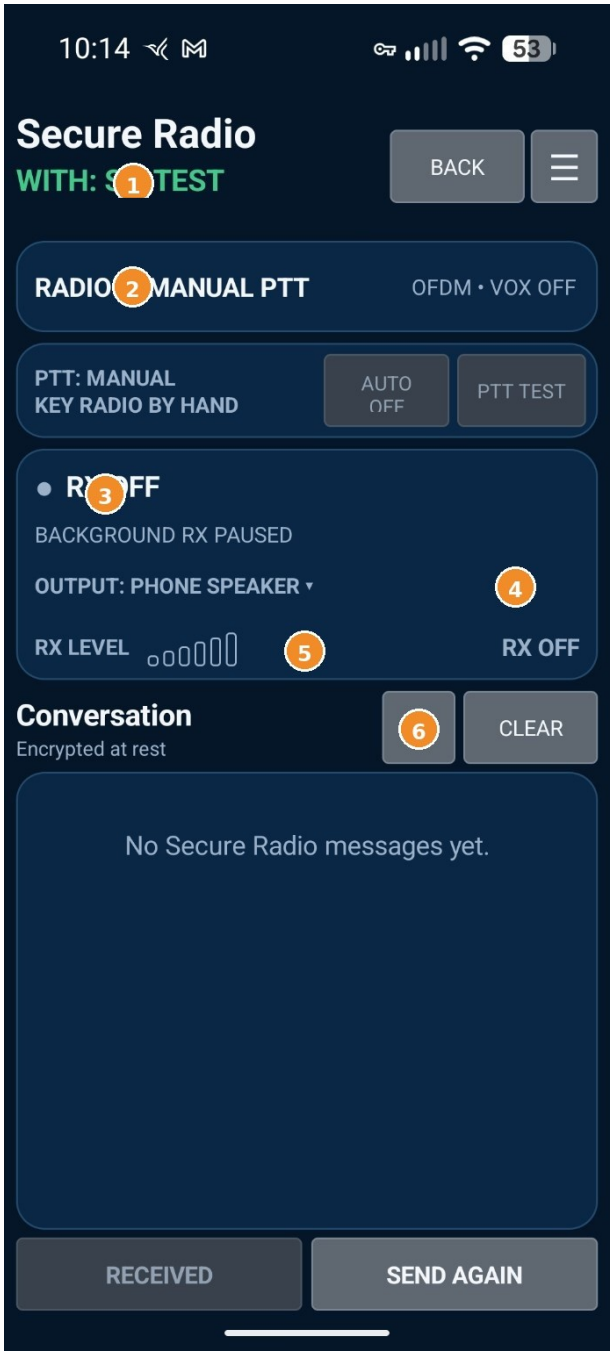
SecureWave uses bounded streaming/chunked authenticated processing for large attachment receive/import. The change was made after a roughly 110 MB attachment exposed a whole-file memory-allocation failure in the prior receive path.

Property	Current behavior
Source file ceiling	199,000,000 bytes
Encrypted transport ceiling	200,000,000 bytes
Chunk size <= 1 MiB files	64 KiB
Chunk size > 1 MiB files	256 KiB
Encryption	AES-256-GCM authenticated records
External handoff	Random .bin name / application/octet-stream

Qualification: The streaming receive/import path is designed to avoid whole-file heap pressure and passed a 110,000,000-byte host round trip under a 64 MB Java heap. In-app viewing and some explicit plaintext export/read paths have separate safety limits and should not be assumed field-qualified at the full 199 MB ceiling until tested on the phone.

8. Secure Radio - RX, Output and PTT

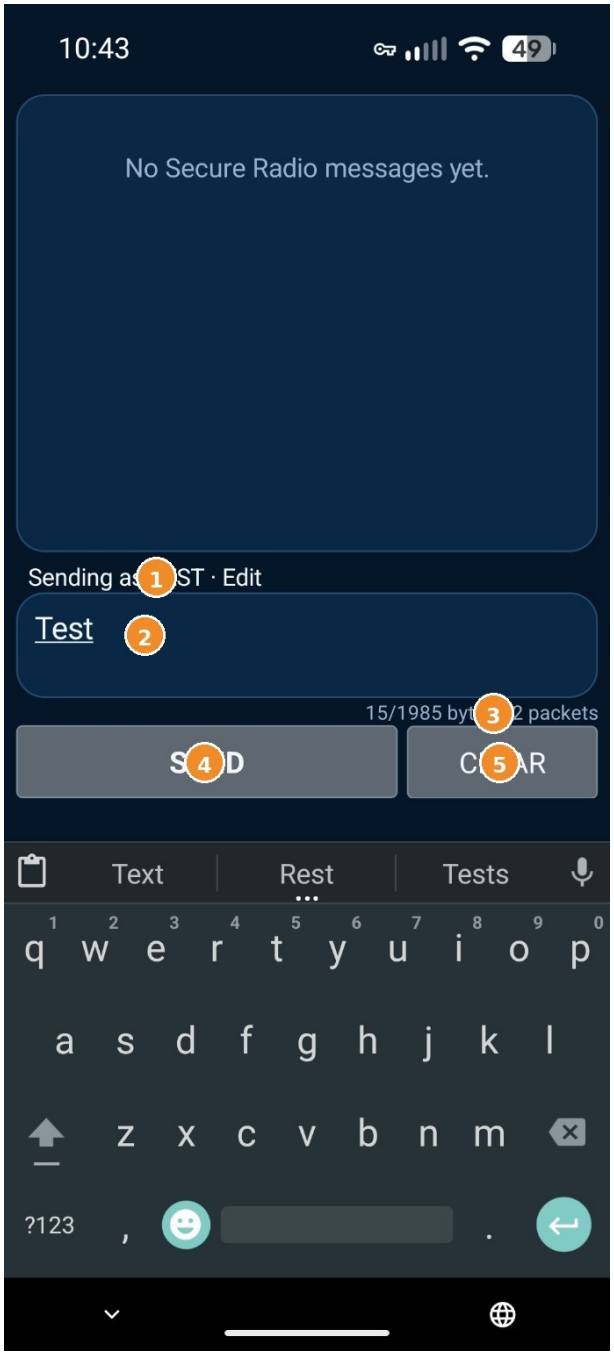
Secure Radio carries encrypted radio messages for the selected secure contact.



1. WITH identifies the secure contact.
2. The top status card identifies RADIO mode, PTT mode, OFDM and VOX state.
3. RX status is a hard receive/privacy control; RX OFF releases receive capture.
4. OUTPUT selects the transmit audio route.
5. RX LEVEL is an audio-level indicator, not a calibrated RF signal-strength meter.
6. Conversation history is encrypted at rest.

8.1 Composing a Secure Radio Message

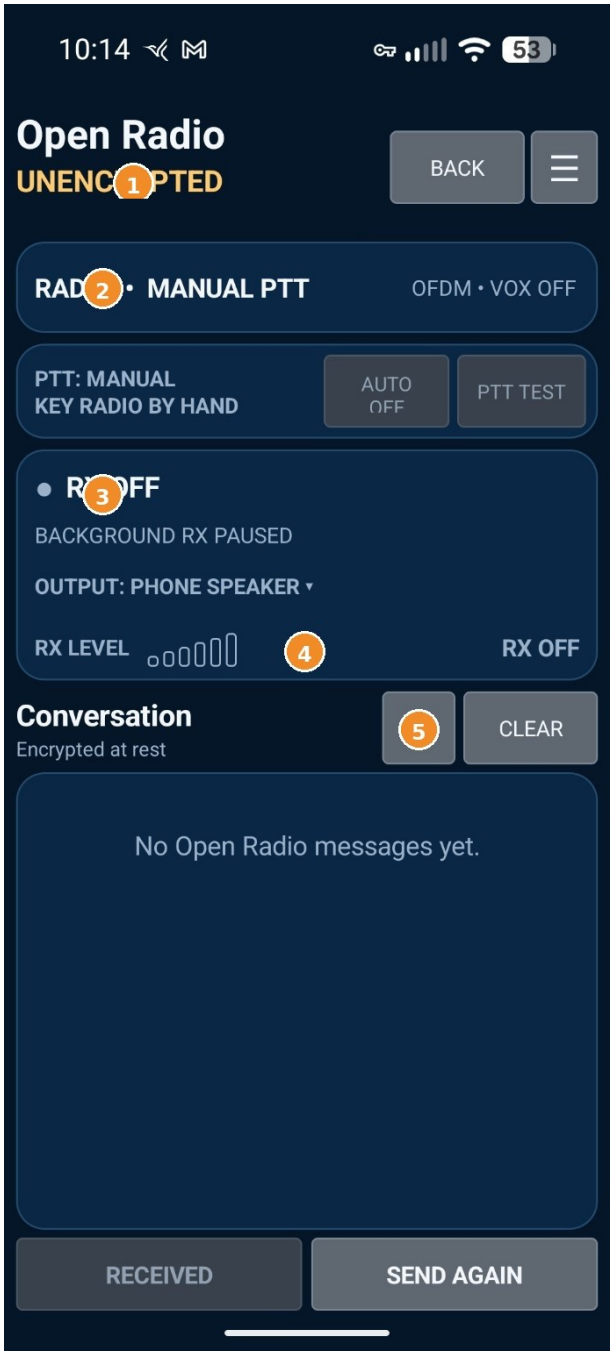
The lower radio screen contains call-sign identity, the scrollable composer and packet estimate.



1. Sending as shows the optional radio call sign and provides Edit.
2. The composer accepts the message and can scroll for longer drafts.
3. The byte/packet counter estimates current radio payload usage.
4. SEND starts the secure radio transmission.
5. CLEAR clears the current draft.

9. Open Radio - Overview

Open Radio uses the same radio transport UI but intentionally does not provide Secure Radio confidentiality or authenticated sender identity.



1. UNENCRYPTED is displayed prominently.

2. Radio/PTT/OFDM status is shown the same way as Secure Radio.

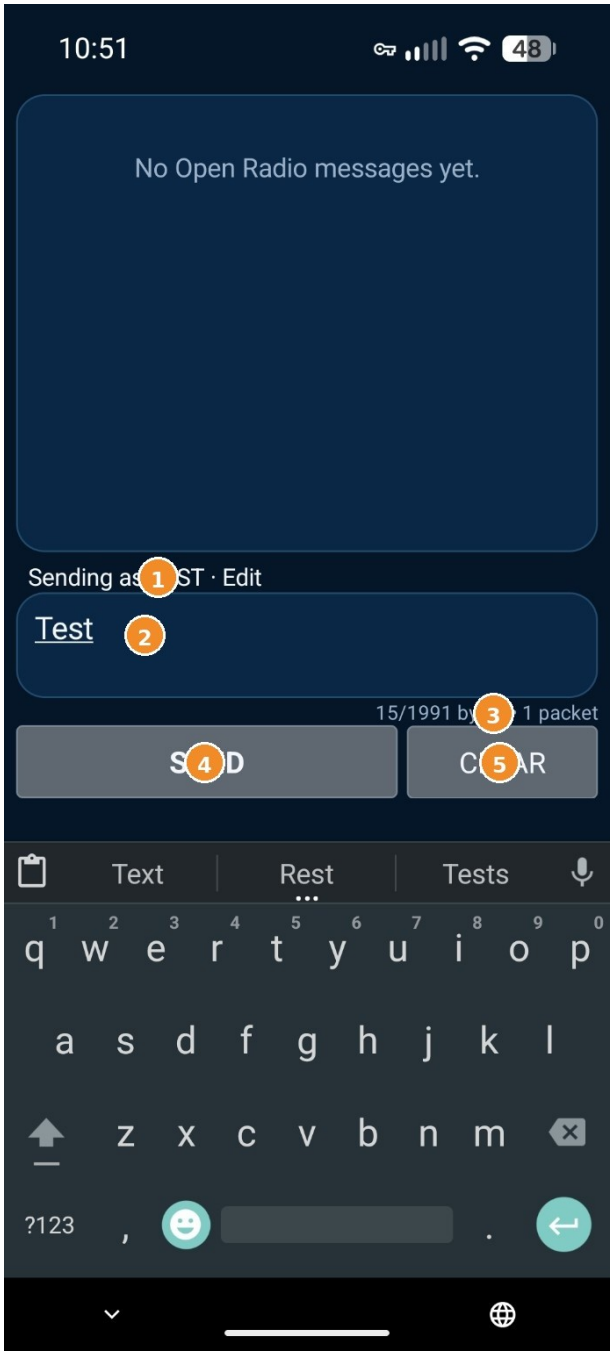
3. RX status and audio route remain operational controls.

4. Conversation history is stored encrypted at rest locally, even though the over-the-air message is unencrypted.

5. CLEAR removes visible conversation history as provided by the UI.

9.1 Composing an Open Radio Message

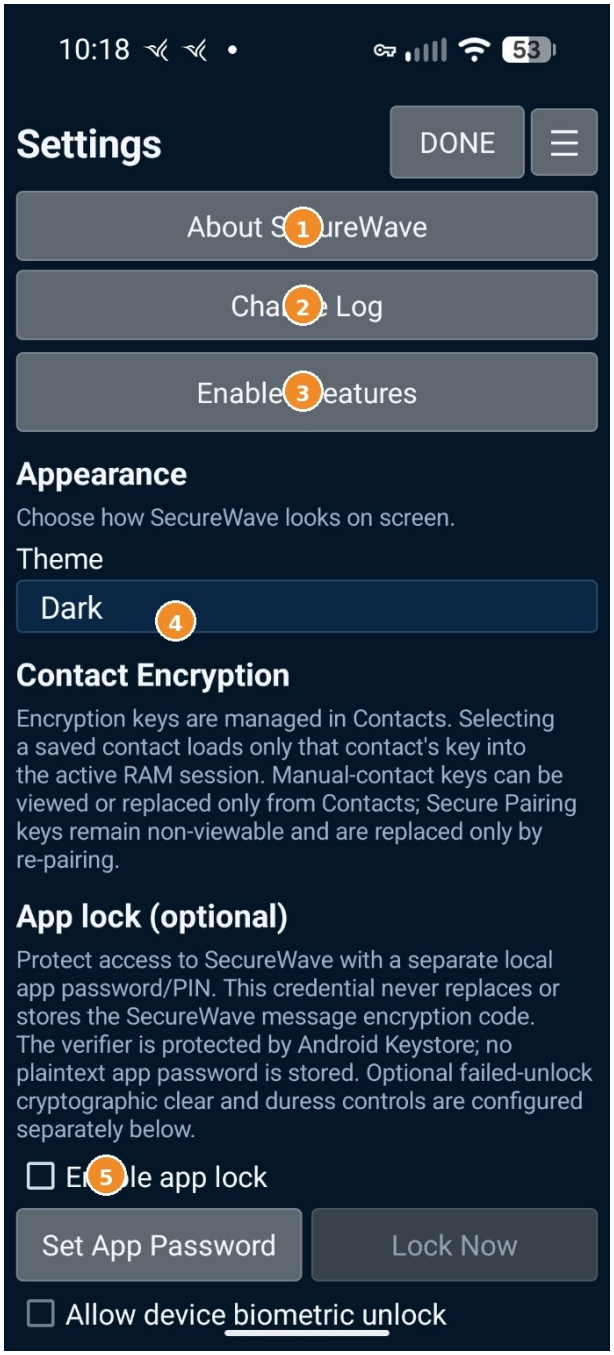
The composer resembles Secure Radio, but the over-the-air payload is unencrypted.



1. Optional call sign.
2. Message composer.
3. Byte/packet counter.
4. SEND transmits the Open Radio message.
5. CLEAR clears the draft.

10. Settings

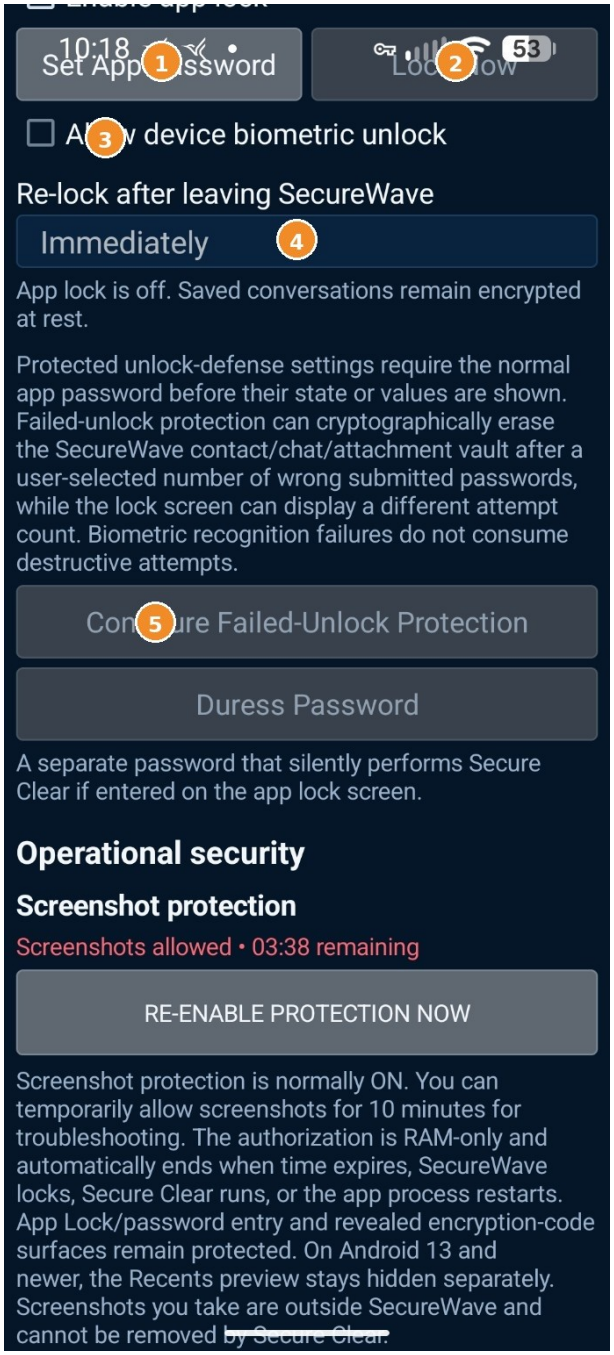
Settings controls appearance, security behavior, App Lock, temporary screenshot permission, and radio preferences. Some illustrated controls have moved since the example screenshot was captured.



1. About SecureWave.
2. Change Log.
3. **Security and radio controls (location may differ by version).**
4. Theme / appearance.
5. App Lock setup and security controls continue lower on the page.

10.1 App Lock Settings

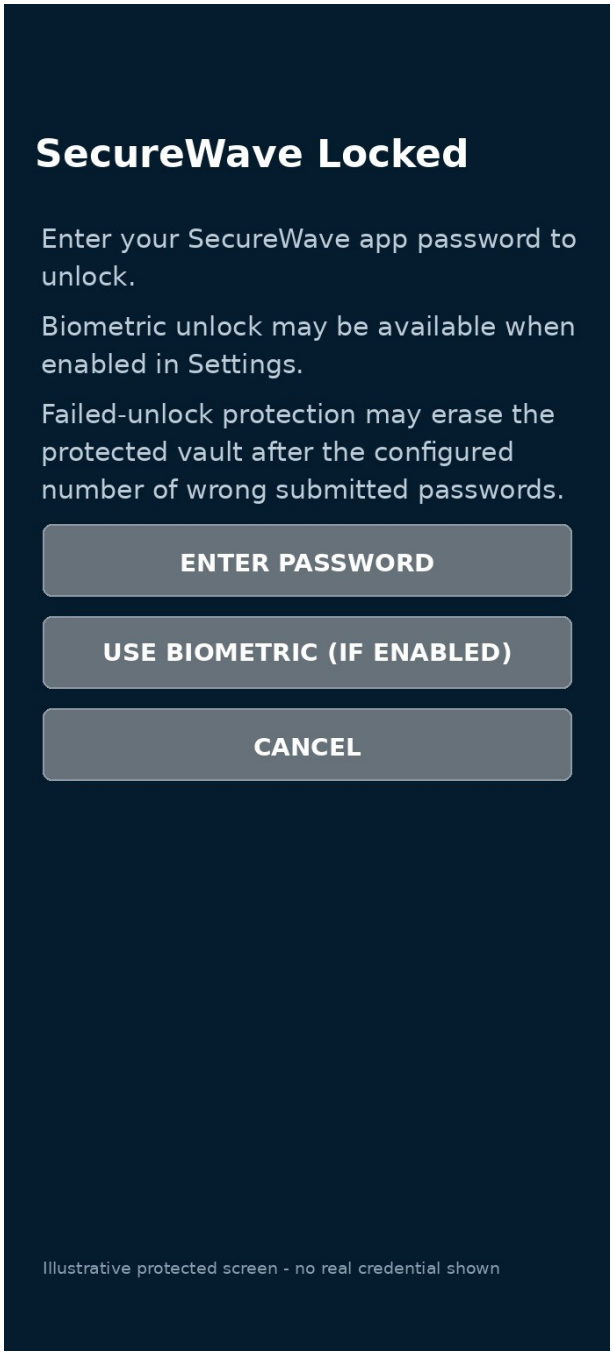
App Lock is optional and is separate from message encryption.



1. Set App Password.
2. Lock Now.
3. Optional device biometric unlock.
4. Re-lock timing after leaving SecureWave.
5. Failed-unlock protection can be configured after the normal app password is established.

10.2 Protected Unlock Screen

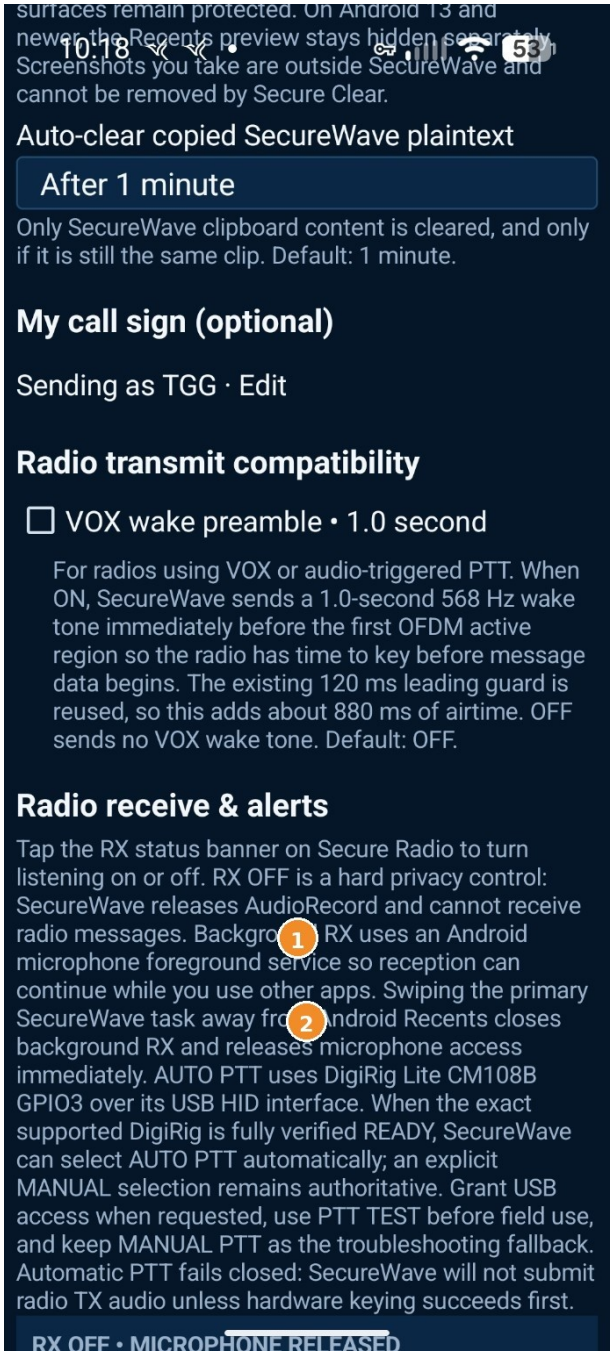
The actual App Lock screen intentionally blocks screenshots. This illustrative reconstruction shows the expected workflow without exposing a real credential.



1. Enter the SecureWave app password.
2. Use biometric unlock only when enabled.
3. Failed-unlock protection may apply to wrong submitted passwords.
4. No real password or sensitive value is shown in this manual.

10.3 Screenshot Protection

Screenshot protection is normally enabled and can be temporarily authorized for troubleshooting.

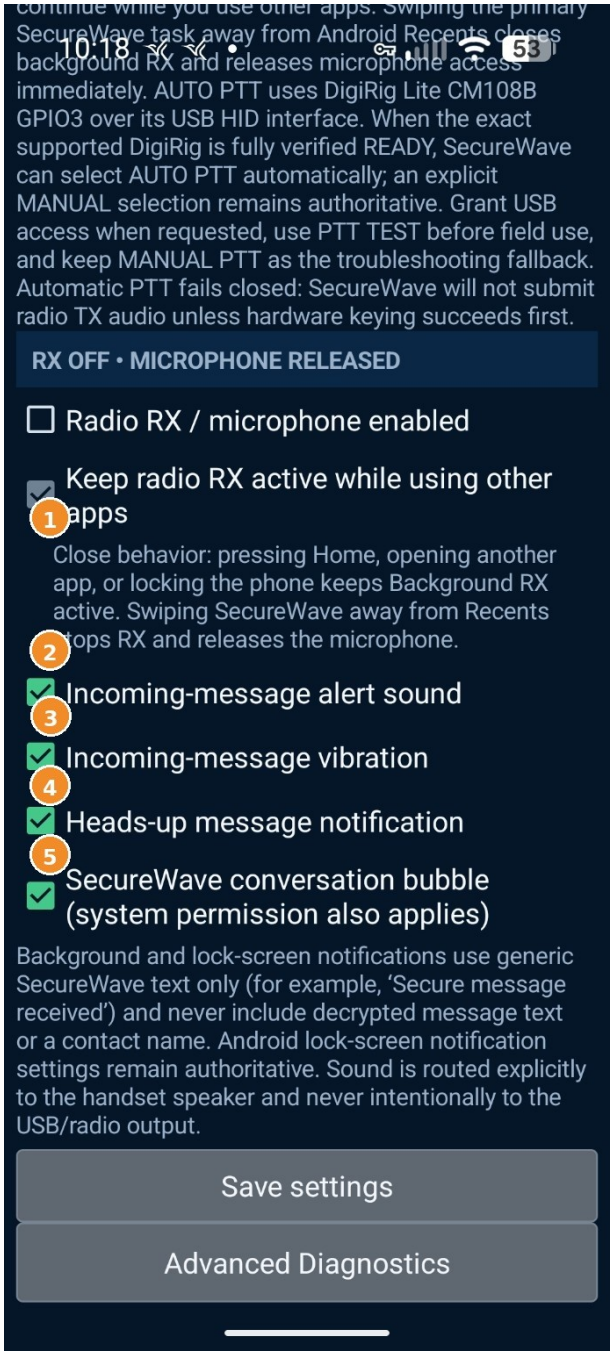


1. The countdown shows how much temporary screenshot authorization remains.

2. RE-ENABLE PROTECTION NOW immediately restores screenshot blocking.

10.4 Radio Receive and Alerts

Settings separates receive privacy, background RX and user alerts.



1. Radio RX / microphone enabled is the primary receive control.
2. Background RX can keep reception active while other apps are in use.
3. Incoming-message alert sound.
4. Incoming-message vibration.
5. Heads-up notification / conversation bubble controls continue below.

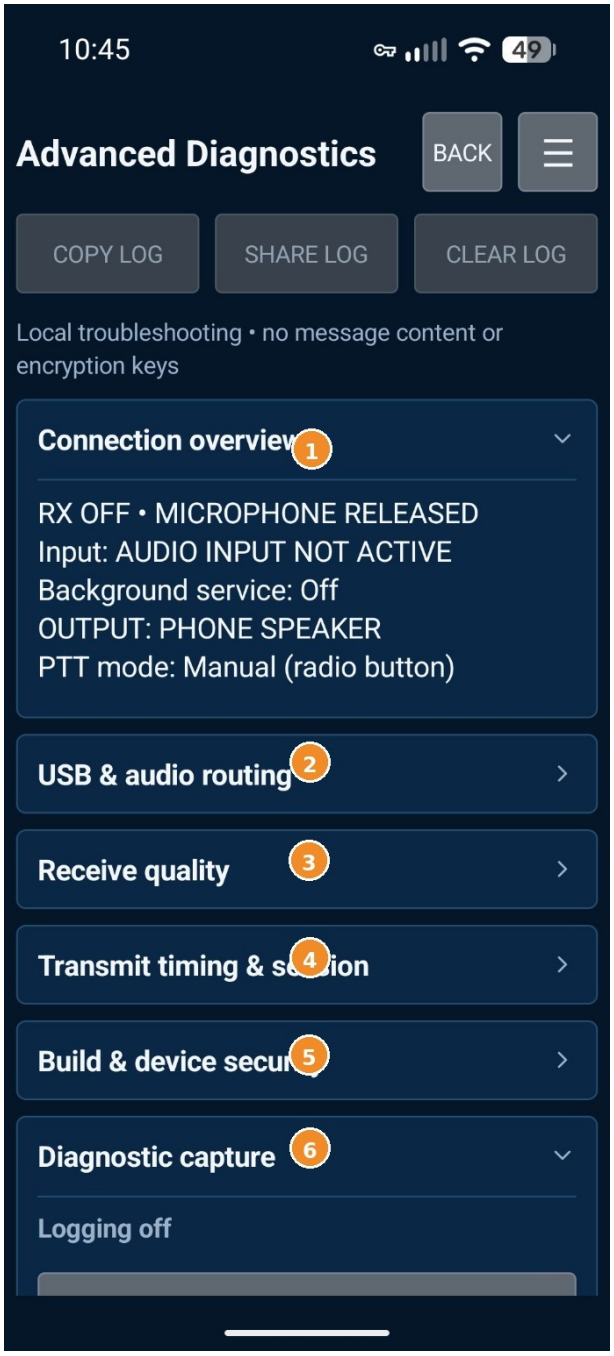
10.5 Offline Operation and Privacy

SecureWave operates without requesting Internet access. Text/file payloads are exported or shared only by a user action; the selected external app may use the Internet and may expose transport metadata. SecureWave itself does not upload content or telemetry to its developer.

Temporary screenshot permission applies to ordinary screens; protected secret-entry screens and the Android Recents preview can remain blocked. Power + Volume Down is the standard Pixel screenshot method. Avoid capturing sensitive contact or pairing material.

11. Advanced Diagnostics

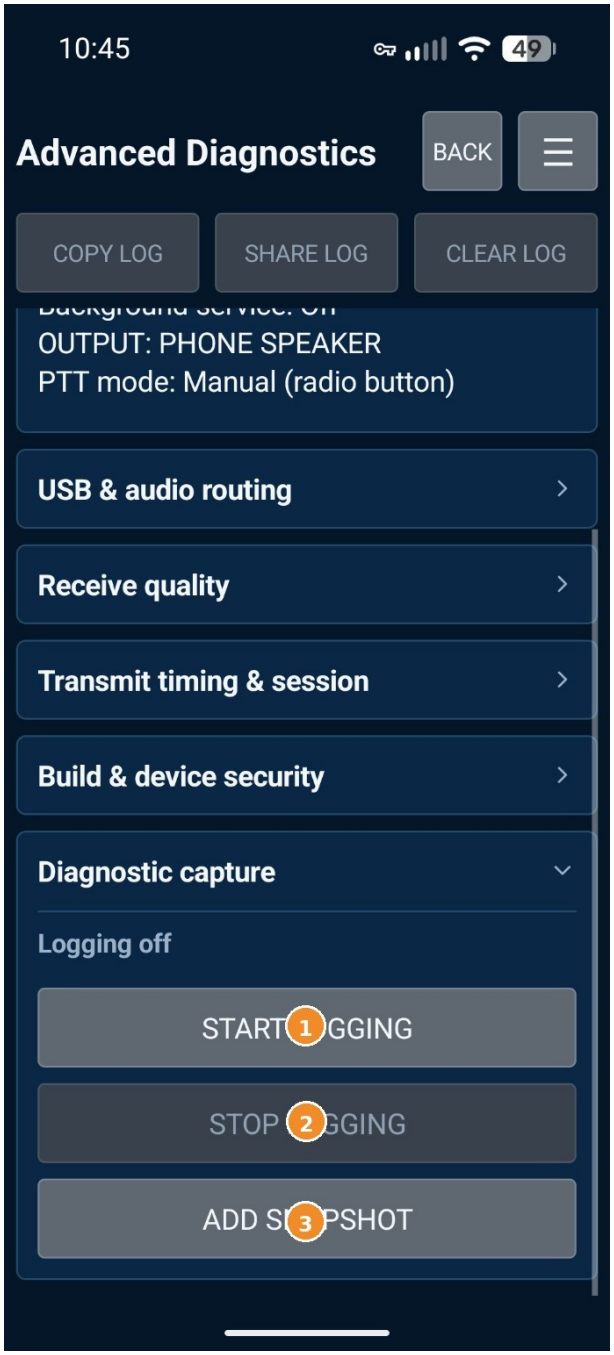
Advanced Diagnostics provides local troubleshooting without intentionally logging message content or encryption keys.



1. Connection overview summarizes RX, input, background service, output and PTT.
2. USB & audio routing.
3. Receive quality.
4. Transmit timing & session.
5. Build & device security.
6. Diagnostic capture.

11.1 Diagnostic Capture

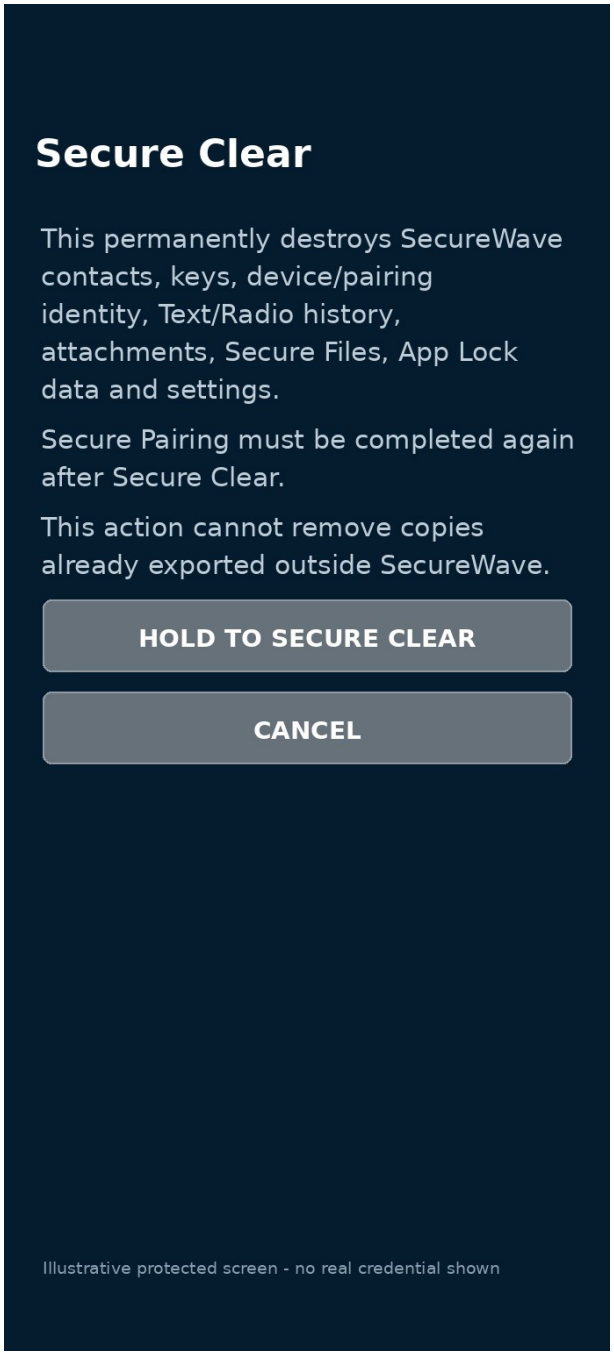
Use logging for a controlled reproduction of a problem.



1. START LOGGING begins capture.
2. STOP LOGGING ends the capture.
3. ADD SNAPSHOT records a diagnostic point during the active investigation.

12. Secure Clear

The Secure Clear confirmation is intentionally protected from screenshots. This reconstruction documents the workflow without weakening that protection.



1. Read the destructive-action warning.
2. Secure Clear destroys contacts, keys, pairing identity, histories, attachments, Secure Files, App Lock data and settings.
3. The action requires deliberate hold/confirmation.
4. Cancel leaves the current SecureWave state intact.

13. Security and Privacy Notes

Internet access and developer collection: SecureWave does not request the Android INTERNET permission. It does not use an account, advertising or analytics SDK, or automatically upload messages, keys, contacts, diagnostics, or usage information to its developer. It processes and stores necessary protected information locally. External messaging apps selected to carry ciphertext have their own permissions and metadata practices.

Contact separation: Securely paired contacts use separate contact-specific cryptographic material. Paired transport derivation is separated by purpose, direction and key epoch.

Secure Text: Secure Text uses authenticated encryption and an opaque padded external representation. The visible transport does not carry the former SW1 prefix.

Hybrid pairing: Hybrid pairing combines P-256 and ML-KEM-768 for key establishment. Identity authentication in this revision remains ECDSA P-256/SHA-256, so the app does not describe the pairing as fully post-quantum authenticated.

Fast Radio Pairing: Fast Radio Pairing is P-256 based, supports verification-code comparison and safe manual resend, and is explicitly not post-quantum.

Attachments: Opaque encrypted file transports use neutral .bin names and generic binary MIME type. Original filename, MIME type, exact size metadata and content are protected inside authenticated encryption.

Metadata limits: Encryption does not hide all metadata. Timing, radio frequency/channel use, the existence of communication and approximate encrypted object sizes can still be observable.

Open Radio: Open Radio is unencrypted over the air even though its local history is encrypted at rest.

Public beta: SecureWave is undergoing field testing. This guide is not an independent cryptographic audit or security certification.

14. FCC Radio Encryption Rules (United States)

SecureWave can generate encrypted radio payloads, but it does not determine whether an RF transmission is lawful. The licensee and operator are responsible for the radio service, frequency, emission, equipment configuration, station identification, and operating practice.

14.1 Amateur Radio Service - Part 97

Do not transmit SecureWave encrypted payloads on United States amateur-radio frequencies. 47 CFR § 97.113(a)(4) prohibits amateur stations from transmitting messages encoded for the purpose of obscuring their meaning. SecureWave encryption intentionally conceals message content and therefore should be treated as prohibited on amateur bands.

Authorized digital modes under 47 CFR § 97.309 do not override the prohibition on obscuring message meaning. Open Radio is unencrypted, but its use on amateur frequencies must still comply with all applicable Part 97 rules, including emission, identification, content, and operator-privilege requirements.

14.2 Part 90 Industrial/Business and Other Licensed Systems

Encryption or scrambling may be permissible on properly licensed Part 90 systems, including some Industrial/Business Pool operations, but a “business frequency” is not blanket permission to encrypt. The licensee must be authorized for the frequency, emission, and type of operation.

47 CFR § 90.212 addresses scrambling and digital voice modulation and ties their use to applicable emission and station-identification requirements. Other Part 90 bands and subparts can impose additional interoperability, encryption, or technical restrictions.

Before using SecureWave, confirm that the station authorization and equipment configuration permit the actual transmitted emission and data use - not merely that the frequency is licensed to a business.

14.3 GMRS and Other Radio Services

Do not assume encryption is permitted merely because a frequency is outside the amateur bands. 47 CFR § 95.1733(a)(3) prohibits GMRS stations from transmitting coded messages or messages with hidden meanings. Other radio services have their own authorization and operating rules.

14.4 Operator Responsibility

SecureWave provides cryptographic capability, not authority to transmit encrypted traffic. Before using Secure Radio, verify current FCC rules, the station license and emission authorization, equipment requirements, and any service-specific restrictions. Outside the United States, follow the applicable national regulator. If uncertain, use a transport clearly authorized for encrypted data or consult the licensing authority.

FCC references: 47 CFR §§ 97.113(a)(4), 97.309, 90.35, 90.212, and 95.1733(a)(3). Rules can change; check the current FCC rules before transmitting.

15. Troubleshooting

Pairing request/response not received

For Fast Radio Pairing, use the relevant SEND ... AGAIN action. Retries reuse the same signed/authenticated stage and can recover missing fragments. For Hybrid pairing, resend/share the complete pairing object without editing it.

Verification codes differ

Cancel the pairing. Do not save the contact.

Secure Text will not decrypt

Confirm the correct contact, complete encrypted payload, and compatible SecureWave release. Obsolete lab formats may be incompatible.

Encrypted file will not import

Confirm the correct contact and complete file. A tampered or incomplete authenticated record is rejected rather than exposed as a partial trusted attachment.

Large file import fails

Confirm current build, free storage and external transport completion. The current release addresses large receive/import heap pressure, but viewing/export paths have separate limits.

Radio receive not working

Confirm RX ON, actual input source, USB/audio routing, correct channel, cable/interface state and that the other radio is transmitting. PHONE MIC means the handset microphone is selected.

AUTO PTT unavailable

Use Manual PTT as the troubleshooting fallback. AUTO requires the exact supported controller to be verified ready.

16. Best Practices

1. Prefer Hybrid Secure Pairing when practical.

2. Compare every verification-code group before saving a contact.

3. Confirm the selected contact before sending encrypted text, radio messages or files.

4. Use strong generated manual keys if manual mode is necessary.

5. Confirm RX, output route and PTT before important radio use.

6. Keep Open Radio and Secure Radio use clearly separated.

7. Treat SAVE DECRYPTED COPY as an intentional export of plaintext from the vault.

8. Use diagnostic logging only for controlled troubleshooting and stop it afterward.

9. Use Secure Clear deliberately; it is a destructive factory-reset-style operation for SecureWave state.

10. Keep both endpoints on compatible SecureWave releases during beta testing.

11. Before using Secure Radio, confirm that encrypted traffic and the actual transmitted emission are permitted on the selected radio service and frequency.

Glossary

AES-256-GCM: Authenticated encryption used for protected SecureWave content.

Hybrid Secure Pairing: The recommended P-256 + ML-KEM pairing method.

Fast Radio Pairing: Compact P-256 pairing carried over radio; not post-quantum.

SAS / verification code: Human-compared short authentication string used to confirm both pairing endpoints derived the same relationship.

StrongBox: Android hardware-backed Keystore implementation when available.

TEE: Trusted Execution Environment; accepted hardware backing where StrongBox is unavailable.

RX / TX: Receive / transmit.

PTT: Push to talk.

OFDM: The modem waveform used for SecureWave radio transport.

Secure Clear: Destructive reset of protected SecureWave state.

Opaque transport: Encrypted external representation designed to avoid obvious SecureWave-specific visible markers.